AGC
Your Dreams, Our Challenge

General Processing Guidelines

N5000

BT Epoxy

Purpose

These guidelines can provide only basic and reference information for PCB fabricators. Because of different environment, equipment, tooling, etc., the user shall determine suitability in any given conditions or applications. For more detailed processing information, please contact your local AGC technical service engineer or sales representative.

Material Handling & Storage

Store laminates flat in a dry environment. Do not bend, scratch, or dent laminate. Store prepreg flat, in a cool, dry environment per IPC-4101E Condition 2 [less than 72°F (23°C) and ≤50% RH]. For extended shelf life, store per IPC-4101E Condition 1 [less than 5°C (41°F)]. Reseal opened bags of unused prepreg.

Copper & Surface Preparation

Prepare copper surface for photo resist application using standard chemistries and techniques.

Bond Enhancing Treatments

One of the following options can be used successfully:

- Option 1: Oxide alternative based on sulfuric-peroxide etch.
- Option 2: Brown oxide with controlled dissolution post-treatment.
- Option 3: Modified brown oxide with DMAB (dimethylamino borane) reduction. To use DMAB successfully, oxide weight gain should be .12 ± .03 mg / cm2.

Note: The brown oxide deposit should be tested using a weight loss test. Thick oxide deposits tend to yield poor thermal resistance. The oxide deposit should not exceed 0.4 mg/cm2.

Inner Layer & Sub-assembly Drying

Inner layers should be oven dried to remove adsorbed moisture prior to re-lamination. Adsorbed moisture in the inner layer can affect the curing properties of the prepreg. Conveyorized warm air drying is usually not effective in removing adsorbed moisture from the etched layer. Consult your oxide supplier for inner layer baking conditions so as to not reduce the intended adhesion properties.

	Best Practice Recommendations
Signal layers	230°F (110°C) in vertical racks with minimum 0.5” (12mm) separation for 60 minutes
Plane layers and sub-assemblies	230°F (110°C) in vertical racks with minimum 0.5” (12mm) separation for 60 minutes

Inner Layer Drying Notes:
For best results, use inner layers within 2 hours after drying. Rebake inner layers if not used within 24 hours. Baking cores in stacks does not provide an effective airflow to remove entrapped moisture from the cores and should be avoided. Post oxide baking is also recommended for each sub-assembly before relamination. The same recommendations outlined above should be followed. Cores and sub-assemblies greater than 100 mils (2.54mm) thick may require additional baking time.

Lamination

For best results, fully cure in vacuum assisted hydraulic press.

	Recommendations
Vacuum Gauge Pressure	A minimum of 28.5" Hg (965 mbars) for 15 minutes. Do not apply heat or pressure prior to vacuum dwell time. Maintain vacuum throughout press cycle.
Heat Up Rate[1]	5 - 12°F (3 - 7°C) per minute
Critical Range	180 - 280°F (80 - 140°C)
Pressure[2]	200 - 350 psi (14 - 24 bar)
Single / Dual Stage	Single Stage: Apply pressure immediately after vacuum dwell time. Dual Stage: Apply full pressure after vacuum dwell time and prior to 150°F (65°C).
Cure Time, Temp	>90 minutes @ 375°F (190°C)
Cool Down Rate	<7°F (4°C) per minute or less until stack reaches 260°F (127°C)
Breakdown	After panels have cooled below 150°F (65 °C)

Lamination Notes:
[1] Heat rise is usually controlled by using an acceptable thermal lagging such as kraft paper or press pads. Alternately the heat rise can be controlled by ramping the platen temperature about 5 - 10°F (5°C) higher than book temperatures and controlling the heat up rate through the critical temperature range.
[2] Lamination pressure is based on 1/2oz and 1oz copper. Heavier copper weights may require higher pressure.

Drilling

Typical Drill Parameters	Recommendations	
Drill Sizes	0.010" - 0.020" (0.30 - 0.50 mm)	0.020" - 0.040" (0.50 - 1.0 mm)
Surface Speed	375 - 450 SFM (115 - 137 m/min)	400-500 SFM (122 - 152 m/min)
Chip Load	0.5 - 1.5 mils/rev (12 - 38 µm/rev)	1.0 - 2.0 mils/rev (25 - 51 µm/rev)
Maximum Hit Count	800 - 1200	1200 - 1500
Typical Stack Height	0.120" - 0.180" (3.0 - 4.5 mm)	0.120" - 0.180" (3.0 - 4.5 mm)

Drilling Notes:
Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, copper content and stack height. For specific feed and speed parameters, contact your drill supplier or AGC's technical representative. Detailed typical drilling parameters are available for many products. Please contact www.agc-ml.info-maltimaterial@agc.com.

Hole Cleaning (Resin Smear Removal)

Most commonly used desmear methods are compatible. Plasma desmear, followed by a short immersion in a permanganate oxidizer is the preferred method for resin smear removal and etch back of hole walls.

Plasma Desmear: Typical desmear conditions

Temperature	Gas mixture	Power	Time
80± 2°C	10%CF_4 , 80% O_2, 10% N_2	4000 W	25 - 30 mins

Plasma Note: Depending on the amount of resin removal required, a preheat cycle and an oxygen burn cycle for ash removal may be necessary. Due to variations in plasma equipment, process times and equipment settings should be evaluated.

Chemical Desmear:

A solvent swell and permanganate oxidizer can also be used for removal of drill smear. This method is not recommended for etch-back as aggressive attack of the resin can result in damage to the hole wall.

Type	Temperature	Time
Cyclic Amine 100%	130 ± 5°F (54 ± 2°C)	5 - 7 mins
Cyclic Amine 50% v/v	170 ± 5°F (77 ± 2°C)	5 - 7 mins
Butyl / hydroxide solvent	173 ± 5°F (78 ± 2°C)	5 - 7 mins
Alkaline Permanganate oxidizer	175 ± 5°F (79 ± 2°C)	12 - 15 mins

Routing

Typical Drill Parameters	Recommendations
Stack Height	0.250” (6 mm)
Tool Size	0.093” (2.4 mm)
Feed Rate	60 IPM (1.5 m/min)
Speed	24K RPM

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